

## M-14

### Automatic AC Transmission Outages Initiated by Failed AC Substation Equipment

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|------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------|------------------------------------------|
| <b>Submittal Date</b>                          | Submitted December 1, 2014                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                              |                                          |
| <b>Proposal Type</b>                           | New <input type="checkbox"/>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Revision <input checked="" type="checkbox"/> | Metric Analysis <input type="checkbox"/> |
| <b>Definition</b>                              | Normalized count on a per transmission element basis (ac circuit or transformer) of 100 kV and above ac transmission element sustained automatic outages initiated by Failed AC Substation Equipment.                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                              |                                          |
| <b>Rating Criteria</b>                         | <ul style="list-style-type: none"> <li>• <b>Red (actionable)</b>: The outage frequency statistically significantly increased compared to the previous four years.</li> <li>• <b>Yellow (monitor)</b>: The outage frequency increased compared to the previous four years, but the increase is not statistically significant.</li> <li>• <b>White (stable)</b>: The outage frequency did not change or decrease compared to the previous four years, and the decrease is not statistically significant.</li> <li>• <b>Green (good/improving)</b>: The outage frequency statistically significantly decreased compared to the previous four years.</li> </ul> |                                              |                                          |
| <b>Purpose</b>                                 | Track Failed AC Substation Equipment as a contributing factor of Sustained Automatic Transmission Outages.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                              |                                          |
| <b>Formula or Type of Statistical Analysis</b> | 100 kV and above Sustained AC Outages initiated by Failed AC Substation Equipment = Number of Sustained Automatic AC Element Outages occurring on 100 kV and above elements initiated by Failed AC Substation Equipment / Total Number of 100 kV and above AC Elements (AC Circuits or Transformers).                                                                                                                                                                                                                                                                                                                                                       |                                              |                                          |
| <b>Time Horizon</b>                            | Five year rolling average                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                              |                                          |
| <b>Metric Start Time or Baseline</b>           | Historical time frame                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                              |                                          |

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|-----------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|------------|------------|----------|---------------------|
| Data Collection Interval and Roll Up                                                                | The TADS data provides the total number of sustained automatic transmission system outages and the number of outages initiated by Failed AC Substation Equipment <sup>1</sup> for 100 kV and above. |                 |            |            |          |                     |
| Ease of Collection                                                                                  | Data collected via the NERC TADS process.                                                                                                                                                           |                 |            |            |          |                     |
| Aggregation                                                                                         | Results available by normalized counts on a NERC wide basis.                                                                                                                                        |                 |            |            |          |                     |
| Links to NERC Standard                                                                              | None                                                                                                                                                                                                |                 |            |            |          |                     |
| Data Source                                                                                         | The NERC TADS definitions and data.                                                                                                                                                                 |                 |            |            |          |                     |
| Data to be Submitted By                                                                             | Entities submitting NERC TADS data.                                                                                                                                                                 |                 |            |            |          |                     |
| Revision History                                                                                    | Reviewed February 19, 2025<br>Revised March 16, 2022<br>Reviewed October 16, 2019<br>Revised on December 2, 2015                                                                                    |                 |            |            |          |                     |
| PAS and NERC Staff Use                                                                              |                                                                                                                                                                                                     |                 |            |            |          |                     |
| Need for Validation or Pilot                                                                        | No, the data and results reported via the TADS process.                                                                                                                                             |                 |            |            |          |                     |
| SMART Rating<br>PAS SMART rating of proposed metric, metric revision, or new metric analysis method | Total Score                                                                                                                                                                                         | Specific/Simple | Measurable | Attainable | Relevant | Tangible/<br>Timely |
|                                                                                                     | 14                                                                                                                                                                                                  | 3               | 3          | 3          | 3        | 2                   |
| Publications and Documentation                                                                      | State of Reliability Report                                                                                                                                                                         |                 |            |            |          |                     |

<sup>1</sup> TADS Data Reporting Instruction can be viewed at: [Transmission Availability Data System \(TADS\) \(nerc.com\)](https://www.nerc.com/Transmission-Availability-Data-System-(TADS)-(nerc.com))